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1  -- =====
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2  -- EX 603 Assignment 2 – schema.sql
3  -- Theme: <Paw Air Online Booking>
4  -- Author: <Serena Lim>
5  -- Target: PostgreSQL 18.6
6  -- =====
   =====
7  -- Reset. Reverse creation order, so no dependency
   blocks a drop.
8  DROP TABLE IF EXISTS bookings CASCADE;
9  DROP TABLE IF EXISTS flights  CASCADE;
10 DROP TABLE IF EXISTS flight_routes  CASCADE;
11 DROP TABLE IF EXISTS passenger CASCADE;
12 DROP TABLE IF EXISTS airports  CASCADE;
13
14 CREATE TABLE airports (
15     airport_id VARCHAR(10) NOT NULL,
16     airport_name VARCHAR NOT NULL UNIQUE,
17     airport_code VARCHAR NOT NULL,
18     airport_city VARCHAR NOT NULL,
19     airport_country VARCHAR NOT NULL,
20     airport_status VARCHAR NOT NULL,
21     CHECK (airport_status IN ('Operate', '
Closed')),
22     PRIMARY KEY (airport_id)
23 );
24
25 CREATE TABLE passenger (
26     passenger_id VARCHAR(20) NOT NULL, --CONSTRAINT
chk_id_range CHECK (id BETWEEN 1 AND 99999999)
27     passenger_username VARCHAR(10) NOT NULL UNIQUE,
28     passenger_first_name VARCHAR(20) NOT NULL,
29     passenger_middle_name VARCHAR(20),
30     passenger_last_name VARCHAR(20) NOT NULL,
31     passenger_DOB DATE NOT NULL,
32     CHECK (passenger_DOB <=
passenger_enroll_date),
33     passenger_email VARCHAR NOT NULL,
34     CHECK (passenger_email ~ '^[A-Za-z0-9._
%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$'),

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35     passenger_phone INT NOT NULL,
36     passenger_enroll_date DATE NOT NULL,
37     CHECK (passenger_enroll_date <=
CURRENT_DATE),
38     passenger_loyalty_id VARCHAR(20) UNIQUE,
39     passenger_address_street VARCHAR(80) NOT NULL,
40     passenger_address_city VARCHAR(20) NOT NULL,
41     passenger_address_state VARCHAR(20),
42     passenger_address_country VARCHAR(20) NOT NULL,
43     PRIMARY KEY (passenger_id)
44 );
45
46 CREATE TABLE flight_routes (
47     route_id VARCHAR(20) NOT NULL,
48     departure_airport_id VARCHAR NOT NULL,
49     arrival_airport_id VARCHAR NOT NULL,
50     route_distance INT NOT NULL,
51     estimated_duration INT NOT NULL,
52     route_status VARCHAR NOT NULL,
53     CHECK (route_status IN ('Active', '
Pending', 'Suspected')),
54     PRIMARY KEY (route_id),
55     FOREIGN KEY (departure_airport_id) REFERENCES
airports(airport_id) ON DELETE SET NULL,
56     FOREIGN KEY (arrival_airport_id) REFERENCES
airports(airport_id) ON DELETE SET NULL,
57     CHECK (departure_airport_id <>
arrival_airport_id)
58 );
59
60 CREATE TABLE flights (
61     flight_id VARCHAR(20) NOT NULL,
62     route_id VARCHAR(20),
63     flight_number VARCHAR(20) NOT NULL,
64     departure_date TIMESTAMP NOT NULL,
65     arrival_date TIMESTAMP NOT NULL,
66     departure_status VARCHAR NOT NULL,
67     CHECK (departure_status IN ('Scheduled'
, 'Cancelled', 'Delayed', 'Early')),
68     arrival_status VARCHAR NOT NULL,
69     CHECK (arrival_status IN ('Scheduled',

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69 'Cancelled', 'Delayed', 'Early'))),
70     PRIMARY KEY (flight_id),
71     FOREIGN KEY (route_id) REFERENCES
    flight_routes(route_id) ON DELETE SET NULL
72 );
73
74 CREATE TABLE bookings (
75     booking_id INT NOT NULL,
76     passenger_id VARCHAR NOT NULL,
77     flight_id VARCHAR NOT NULL,
78     booking_status VARCHAR NOT NULL,
79     CHECK (booking_status IN ('Confirmed', '
    Cancelled', 'Pending')),
80     booking_date TIMESTAMP NOT NULL,
81     fare_price FLOAT NOT NULL,
82     CHECK (fare_price > 0),
83     PRIMARY KEY (booking_id),
84     FOREIGN KEY (passenger_id) REFERENCES
    passenger(passenger_id) ON DELETE CASCADE,
85     FOREIGN KEY (flight_id) REFERENCES flights(
    flight_id) ON DELETE SET NULL
86 );
87
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